# DPUCreateProcess

This DPU creates and controls an application a computer.

## 1. Static parameters:

| | |
|---|---|
| *ProcessName* | Path of the application (EXE-file). If no directory is given the program is searched for in the system directories (environment variable PATH). |
| *WindowTitle* | Title of the main window of the application to be started. The parameter is used to control the application (e.g. show, hide and terminate). A * can be appended to WindowTitle. In this case the window title is matched only with the prefix before *. This is helpful if the application displays a file name as suffix of the window title. |
| *CurrentDirectory* | Working directory of the application to be started. If not specified the SILAB\bin directory is used. |
| *CommandLine* | Command line arguments of the application to be started. |
| *Environment* | Defines additional environment variables for the application (see below, section „environment variables“). |

**Extensions to the static parameters above:**

The following special tokens allow to include system specific information in the static parameters:

| | |
|---|---|
| *%SILAB%* | SILAB installation directory (e.g. *D:\SILAB*). |
| *%BIN%* | Directory for SILAB binaries (e.g. *D:\SILAB\bin*). |
| *%PROJECT%* | Project path from which the current configuration has been loaded (e.g. *D:\SILAB\Projects\SILABDemo*). |
| *%QUOTE%* | A quotation mark ("). |
| *%VAR%* | Is replaced by the environment variable VAR. |

**Static parameters (actions):**

For each of the following parameters several actions can be defined (see table below):

| | |
|---|---|
| *OnPrepare* | Determines what has to be done during the simulation preparation phase.<br>Default: Start the application |
| *OnReadyToStart* | Determines what has to be done during the launch step of the simulation.<br>Default: Nothing |
| *OnStart* | Determines what has to be done at start of the simulation.<br>Default: Start the application if this has not yet been done during the preparation phase. |
| *OnPause* | Determines what has to be done during pause of the simulation.<br>Default: Nothing |

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| | |
|---|---|
| *OnResume* | Determines what has to be done if the simulation is resumed after pause.<br>Default: Nothing |
| *OnReadyToStop* | Determines what has to be done in the stopping phase of the simulation.<br>Default: Terminate the application if it has been started at simulation start. Otherwise: nothing |
| *OnStop* | Determines what has to be done immediately after stop of the simulation.<br>Default: Nothing |
| *OnShutdown* | Determines what has to be done after termination of the simulation (e.g. after closing SILAB).<br>Default: Terminate the application except *LeaveRunning=true*. |

**Static parameters (controls):**

| | |
|---|---|
| *StartupDelayMS* | Waits the specified amount of time after starting the application in *OnPrepare* or *OnReadyToStart*. This can be used to ensure that DPUs following after DPUCreateProcess can access the application.<br>Default: 0 ms |
| *WindowStyle* | Determines how the application is displayed:<br>0 (Default): The application decides<br>1: Minimized<br>2: As a window<br>3: Maximized |
| *WindowX*, *WindowY*<br>*WindowW*, *WindowH* | Determines position and size of the application window.<br>Default: Don’t change size and position of the window. |
| *LeaveRunning* | If this parameter is *true*, the application runs even after SILAB has been terminated.<br>Default: *false* |
| *Kill* | = *false* (Default): The Windows message *WM_CLOSE* is sent to the application for termination. This corresponds to the message that is normally sent if the menu command File/Close is choosen.<br>= *true*: The process is killed for termination. The application might not be ready and unsaved data might be lost. |

## 2. Environment variables:

The section Environment allows to specify additional environment variables or to adjust previously defined values. The following example shows how to include this section in the DPU definition:

```
DPUCreateProcess Proc {
	…
	Environment Env {
		PATH = "%PATH%;%BIN%";
		SILAB_HOME = "%SILAB%";
	};
};
```

If no such section is defined DPUCreateProcess does the following:

- All environment variables that were defined when SILAB was started are applied.
- For each definition *Var* = *Text*;  within the Environment section, the value of the environment variable *Var* is set to *Text*.

## 3. Functional principle:

The *DPUCreateProcess* can be used to start and control an external application during simulation.

Thereby, in each state of the simulation the actions specified with the parameters *OnXXX* are performed.

The following actions are possible:

| Value | Name | Description |
|---|---|---|
| 0 | do nothing | *DPUCreateProcess* doesn't influence the application in the simulation state. |
| 1 | start | *DPUCreateProcess* starts the application if it has not been started earlier. |
| 2 | terminate | *DPUCreateProcess* terminates the application if it has been started by the same instance of *DPUCreateProcess* within the current simulation run. |
| 3 | show | *DPUCreateProcess* ensures that the main window of the application is visible. It also brings the main window to the foreground. Both only happen if the application has been started already. |
| 4 | hide | *DPUCreateProcess* hides the main window of the application. This only happens if the application has been started already. |
| 5 | foreground | Brings the main window of the application to the foreground. |

The following happens if the action *terminate* is issued:

- If *Kill = false*, *DPUCreateProcess* tries to close the application by sending the Windows message *WM_CLOSE*. This corresponds to pressing Alt+F4 or choosing the menu option File/Close in the application.
- If this standard message is not handled correctly by the application, then it cannot be closed by *DPUCreateProcess*. In this case the application has to be killed by setting *Kill = true*.

**Note**: With the exception of the described actions *DPUCreateProcess* does not communicate with the application. It does not control the application in any other way. Thus, it is important that the application can be terminated and that it does not disturb the simulation (e.g. by using high amounts of processing capacity).

## 4. Example:

Launch a Kanzi application (external HMI tool) with SILAB:

```
DPUCreateProcess CreateKanziProcessCluster
{
	Index = 460;
	Computer = { DSP };
```

```
    ProcessName = "Application Player\Kanzi.exe";
    CurrentDirectory = "Application Player";
    WindowTitle = "Kanzi";
    StartupDelayMS = 0;
    WindowStyle = 3;
    WindowX = 0;
    WindowY = 0;
    WindowW = 3840;
    WindowH = 1080;
    LeaveRunning = false;
    Kill = true;
};
```